Supplementary Materials for

VLSA: Vision-Language-Action Models with Plug-and-Play Safety Constraint Layer

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Supplementary Note S1: Vision-Language Model Prompt for Obstacle Identification

To constrain the VLM reasoning process and ensure the output is strictly formatted as a concise textual grounding query, we design the following zero-shot prompt:

> *"The robot must follow this instruction: [Instruction]. Based on both the instruction and the image, identify exactly one non-robot object that is most likely to obstruct the robot's motion during task execution. You must output a uniquely identifiable obstacle name including both color and object type, preferably from this list when applicable: [List]. Output only the object name, with no additional words."*

Here, `[Instruction]` is dynamically replaced with the specific task command provided to the robot (e.g., *"Pick up the black bowl on the stove and place it on the plate"*). The `[List]` variable contains a predefined set of semantic object categories present in the current workspace, which significantly aids in reducing open-vocabulary hallucination and ensures the generated textual query aligns with the observable environment.

Supplementary Note S2: Action-driven Safety-guaranteed Control Procedure

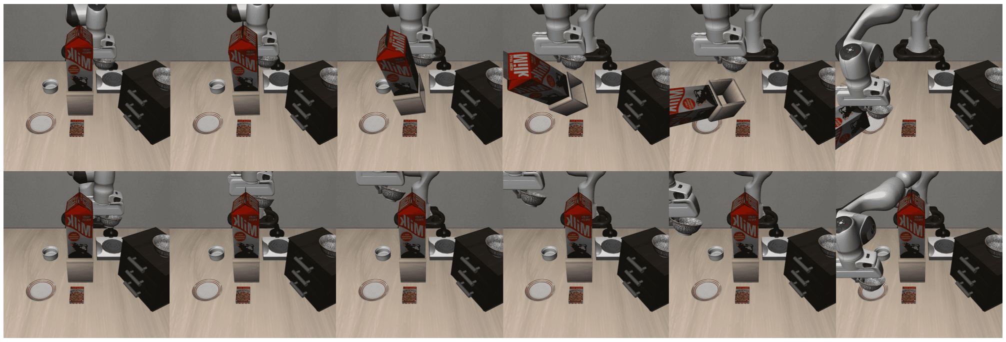
Algorithm 1: Action-driven Safety-guaranteed Control Procedure

Input: Obstacle ellipsoid parameters $\mathcal{E}_{ob} = \{\boldsymbol{p}_{ob}, \boldsymbol{Q}_{ob}, \boldsymbol{R}_{ob}\}$, end-effector geometric parameters $\boldsymbol{Q}_{ef}, \Delta\boldsymbol{p}$, end-effector position $\boldsymbol{p}_{ef}$, rotation matrix $\boldsymbol{R}_{ef}$, VLA policy π_{vla}

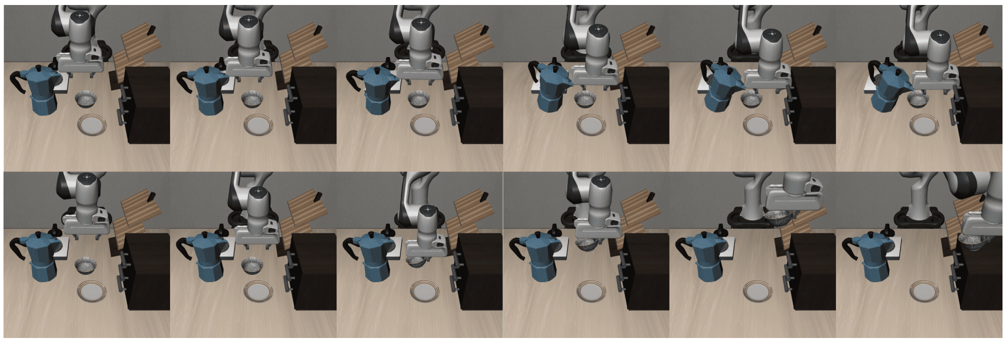
Output: Safe action output $\boldsymbol{u}_{\text{safe}}$

Initialize the virtual state $\boldsymbol{p}_s$ on the unit sphere surface;
while *task not completed* **do**
    Get current observation o_t and language instruction l;
    Construct system state $\boldsymbol{x} \leftarrow [\boldsymbol{p}_{ep}, \boldsymbol{R}_{ef}, \boldsymbol{p}_s]$;
    Generate nominal reference action $\boldsymbol{u}_{\text{vla}} \leftarrow \pi_{\text{vla}}(o_t, l)$;
    Construct barrier function value $h(\boldsymbol{x})$ via Eq. (9);
    Solve the QP defined in Eq. (2) and obtain $\boldsymbol{u}_{\text{safe}}$ and $\boldsymbol{u}_{p_s}$;
    Execute safe action $\boldsymbol{u}_{\text{safe}}$ on the augmented system to update robot states $\boldsymbol{p}_{ef}, \boldsymbol{R}_{ef}$ and virtual state $\boldsymbol{p}_s$;
end

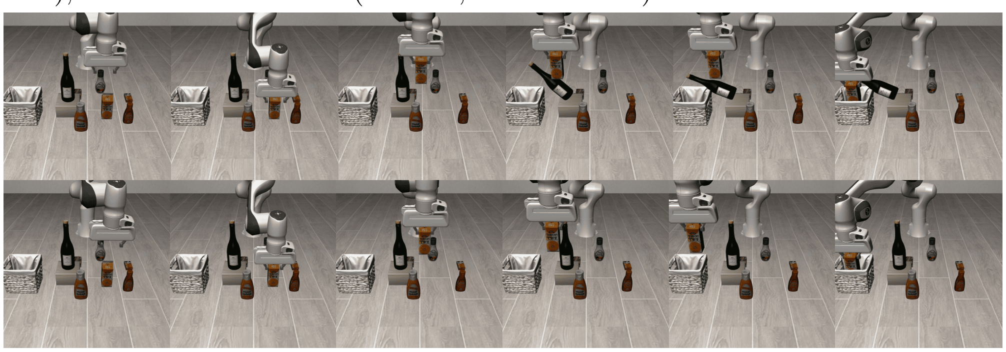
Supplementary Figure S1: Visual Comparison of Task Execution Processes in Simulation Studies



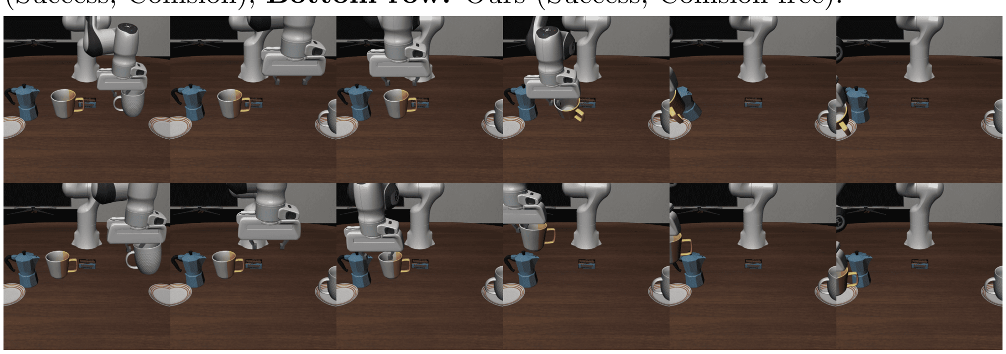
(a) Task: Pick up the bowl on the stove and place it on the plate. **Top row:**$\pi_{0.5}$ (Failure, Collision); **Bottom row:** Ours (Success, Collision-free).



(b) Task: Put the bowl on top of the cabinet. **Top row:** $\pi_{0.5}$ (Failure, Collision); **Bottom row:** Ours (Success, Collision-free).



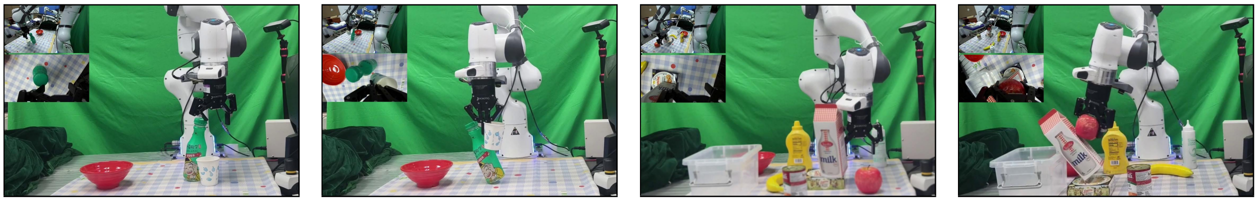
(c) Task: Pick up the orange juice and place it in the basket. **Top row:** $\pi_{0.5}$ (Success, Collision); **Bottom row:** Ours (Success, Collision-free).



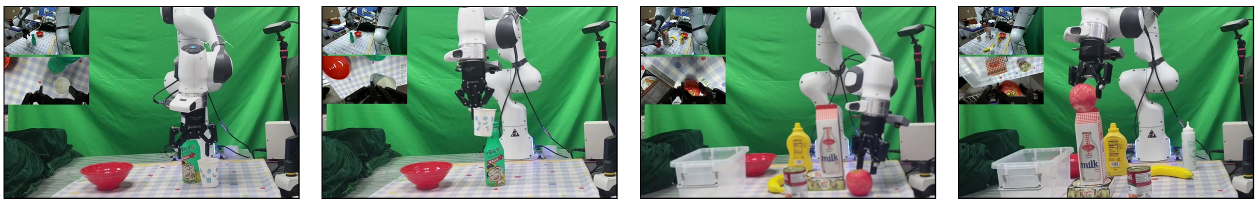
(d) Task: Put the white mug on the right plate and put the yellow and white mug on the left plate. **Top row:** $\pi_{0.5}$ (Success, Collision); **Bottom row:** Ours (Success, Collision-free).

Figure S1: Visual comparison of task execution processes on representative tasks in simulation studies.

Supplementary Figure S2: Visual Comparison of Task Execution Processes in Real-World Experiments



(a) Level I ($\pi_{0.5}$) (b) Level II ($\pi_{0.5}$) (e) Level I ($\pi_{0.5}$) (f) Level II ($\pi_{0.5}$)



(c) Level I (Ours) (d) Level II (Ours) (g) Level I (Ours) (h) Level II (Ours)

Figure S2: Qualitative Comparison. Visualizing actions for the Cup task (Left) and Apple task (Right). For each task, the top row shows the baseline $\pi_{0.5}$ failing to avoid obstacles, while the bottom row shows our method successfully generating collision-free actions.

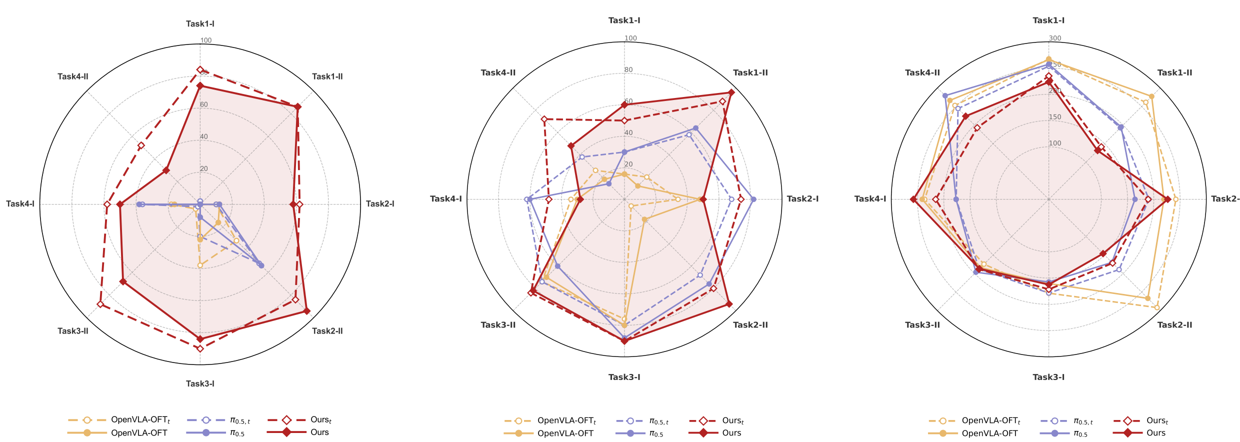
Supplementary Note S3: Details Results of Simulation Studies

Quantitative results for individual tasks across all suites are detailed in Tables S1-S4. For intuitive comparison, we visualize these results as radar charts in Figures S3-S6.

Table S1: **Quantitative results on SafeLIBERO-Spatial suite.** Comparisons under both translational-only and full action space settings. **Bold** indicates the best result (higher is better for CAR/TSR, lower is better for ETS).

Task Description	**Level**	**Translational Action Space Only**									**Full Action Space**								
		OpenVLA-OFT$_t$			$\pi_{0.5,t}$			**Ours$_t$**			OpenVLA-OFT			$\pi_{0.5}$			**Ours**		
		C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓
Pick up the black bowl between the ramekin and the plate and place it in the plate	I	0.0	16.0	267.3	2.0	30.0	253.6	**84.0**	**50.0**	**235.0**	0.0	16.0	265.9	0.0	30.0	257.3	**74.0**	**60.0**	**223.5**
	II	0.0	20.0	261.3	0.0	58.0	192.7	**86.0**	**88.0**	**141.4**	0.0	12.0	277.2	0.0	64.0	194.9	**86.0**	**96.0**	**131.4**
Pick up the black bowl on the cabinet and place it on the plate	I	10.0	34.0	242.0	10.0	68.0	190.6	**62.0**	**74.0**	**189.3**	12.0	48.0	219.3	12.0	**82.0**	**164.2**	**58.0**	50.0	226.6
	II	32.0	6.0	291.6	52.0	68.0	189.3	**84.0**	**80.0**	**171.8**	16.0	18.0	266.8	54.0	76.0	170.0	**94.0**	**94.0**	**146.2**
Pick up the black bowl on the stove and place it on the plate	I	38.0	76.0	178.8	20.0	80.0	178.5	**90.0**	**90.0**	**171.6**	22.0	80.0	160.3	8.0	88.0	**157.5**	**84.0**	**90.0**	162.5
	II	4.0	74.0	**174.6**	2.0	74.0	187.6	**88.0**	**84.0**	188.2	0.0	70.0	**184.1**	0.0	60.0	196.3	**68.0**	**82.0**	187.0
Pick up the black bowl on the ramekin and place it on the plate	I	18.0	34.0	236.4	36.0	**62.0**	**176.4**	**58.0**	48.0	215.5	16.0	30.0	241.1	38.0	**60.0**	**176.6**	**50.0**	28.0	257.9
	II	0.0	26.0	252.8	0.0	38.0	244.5	**52.0**	**72.0**	**192.8**	0.0	18.0	266.4	0.0	14.0	279.2	**30.0**	**48.0**	**223.9**
Average	-	12.8	35.8	238.1	15.3	59.8	201.7	**75.5**	**73.3**	**188.2**	8.3	36.5	235.1	14.0	59.3	199.5	**68.0**	**68.5**	**194.9**

C: CAR (Collision Avoidance Rate); **T**: TSR (Task Success Rate); **E**: ETS (Execution Time Steps).



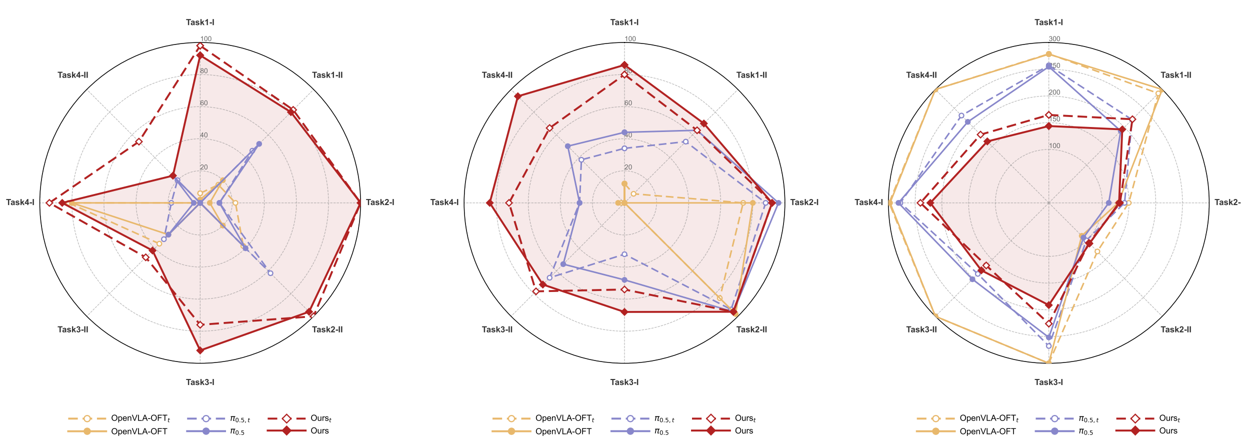
(a) Collision Avoidance Rate (↑) (b) Task Success Rate (↑) (c) Execution Time Steps (↓)

Figure S3: Performance on SafeLIBERO-Spatial suite. Comparison of CAR, TSR, and ETS.

Table S2: **Quantitative results on SafeLIBERO-Goal suite.** Comparisons under both translational-only and full action space settings. **Bold** indicates the best result (higher is better for CAR/TSR, lower is better for ETS).

Task Description	**Level**	**Translational Action Space Only**									**Full Action Space**								
		OpenVLA-OFT$_t$			$\pi_{0.5,t}$			**Ours**$_t$			OpenVLA-OFT			$\pi_{0.5}$			**Ours**		
		C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓
Put the bowl on the plate	I	6.0	12.0	278.7	0.0	34.0	257.1	**98.0**	**80.0**	**164.5**	2.0	12.0	278.1	0.0	44.0	254.7	**92.0**	**86.0**	**143.7**
	II	16.0	8.0	289.6	46.0	54.0	**220.1**	**82.0**	**64.0**	221.2	20.0	0.0	300.0	52.0	64.0	**190.2**	**80.0**	**70.0**	194.3
Put the bowl on top of the cabinet	I	22.0	74.0	149.5	12.0	88.0	141.7	**100.0**	**92.0**	**133.1**	6.0	80.0	129.8	12.0	**96.0**	**112.1**	**100.0**	92.0	131.4
	II	38.0	84.0	128.5	62.0	94.0	**99.3**	**100.0**	**96.0**	106.0	20.0	**98.0**	**87.1**	40.0	96.0	91.8	**96.0**	96.0	107.5
Put the bowl on the stove	I	0.0	0.0	300.0	0.0	32.0	267.7	**76.0**	**54.0**	**225.9**	0.0	0.0	300.0	0.0	48.0	251.6	**92.0**	**68.0**	**191.2**
	II	36.0	0.0	300.0	32.0	66.0	187.8	**48.0**	**78.0**	**165.5**	28.0	0.0	300.0	28.0	54.0	202.0	**42.0**	**72.0**	**178.4**
Open the top drawer and put the bowl inside	I	82.0	2.0	298.7	18.0	28.0	277.6	**94.0**	**72.0**	**240.1**	80.0	4.0	297.0	4.0	28.0	281.2	**86.0**	**84.0**	**221.6**
Put the cream cheese in the bowl	II	0.0	0.0	300.0	20.0	38.0	231.0	**54.0**	**66.0**	**180.6**	0.0	0.0	300.0	**24.0**	50.0	214.3	**24.0**	**94.0**	**162.6**
Average	-	25.0	22.5	255.6	23.8	54.3	210.3	**81.5**	**75.3**	**179.6**	19.5	24.3	249.0	20.0	60.0	199.7	**76.5**	**82.8**	**166.3**

C: CAR (Collision Avoidance Rate); **T**: TSR (Task Success Rate); **E**: ETS (Execution Time Steps).



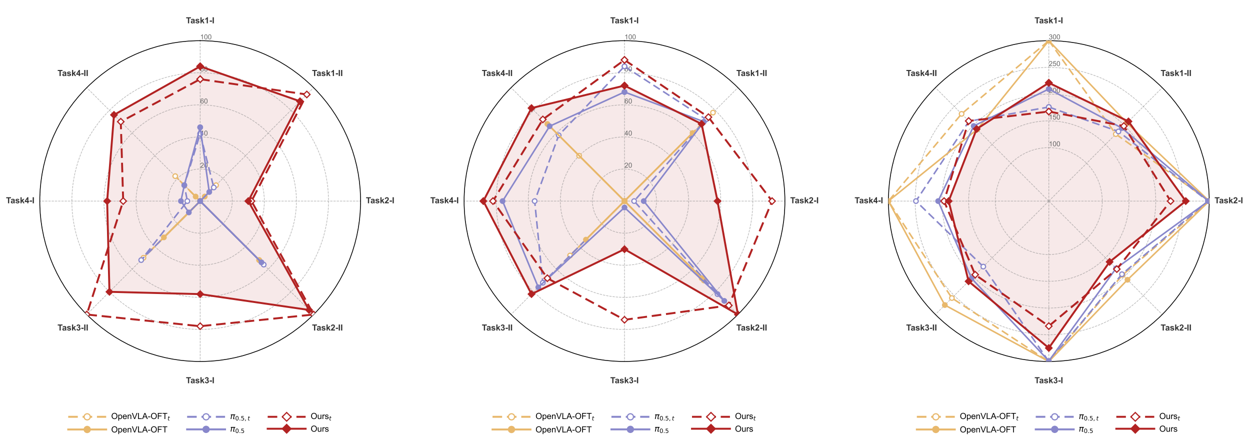
(a) Collision Avoidance Rate (↑) (b) Task Success Rate (↑) (c) Execution Time Steps (↓)

Figure S4: Performance on SafeLIBERO-Goal suite. Comparison of CAR, TSR, and ETS.

Table S3: **Quantitative results on SafeLIBERO-Object suite.** Comparisons under both translational-only and full action space settings. **Bold** indicates the best result (higher is better for CAR/TSR, lower is better for ETS).

Task Description	**Level**	**Translational Action Space Only**									**Full Action Space**								
		OpenVLA-OFT$_t$			$\pi_{0.5,t}$			**Ours$_t$**			OpenVLA-OFT			$\pi_{0.5}$			**Ours**		
		C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓
Pick up the orange juice and place it in the basket	I	0.0	0.0	300.0	42.0	84.0	175.9	**76.0**	**88.0**	**167.5**	0.0	0.0	300.0	46.0	68.0	**209.6**	**84.0**	**72.0**	221.2
	II	14.0	**78.0**	**177.2**	12.0	72.0	184.2	**94.0**	74.0	198.5	4.0	60.0	206.7	8.0	**70.0**	**193.2**	**88.0**	68.0	210.7
Pick up the chocolate pudding and place it in the basket	I	0.0	0.0	300.0	0.0	6.0	297.6	**32.0**	**92.0**	**227.8**	0.0	0.0	300.0	0.0	12.0	298.5	**30.0**	**58.0**	**256.1**
	II	52.0	70.0	195.2	56.0	82.0	193.4	**100.0**	**92.0**	**179.9**	52.0	68.0	208.1	54.0	88.0	177.9	**96.0**	**100.0**	**160.8**
Pick up the milk and place it in the basket	I	0.0	0.0	300.0	0.0	0.0	300.0	**78.0**	**74.0**	**233.8**	0.0	0.0	300.0	0.0	4.0	299.0	**58.0**	**30.0**	**274.8**
	II	50.0	48.0	256.4	52.0	**72.0**	**173.3**	**100.0**	68.0	194.7	32.0	34.0	275.0	10.0	76.0	**204.4**	**80.0**	**82.0**	212.2
Pick up the bbq sauce and place it in the basket	I	0.0	0.0	300.0	8.0	56.0	248.5	**48.0**	**82.0**	**195.9**	0.0	0.0	300.0	12.0	76.0	207.0	**58.0**	**88.0**	**187.3**
	II	22.0	40.0	230.7	14.0	58.0	**211.2**	**70.0**	**72.0**	211.9	4.0	68.0	**189.5**	14.0	66.0	198.6	**76.0**	**82.0**	190.8
Average	-	17.3	29.5	257.4	23.0	53.8	223.0	**74.8**	**80.3**	**201.3**	11.5	28.8	259.9	18.0	57.5	223.5	**71.3**	**72.5**	**214.2**

C: CAR (Collision Avoidance Rate); **T**: TSR (Task Success Rate); **E**: ETS (Execution Time Steps).



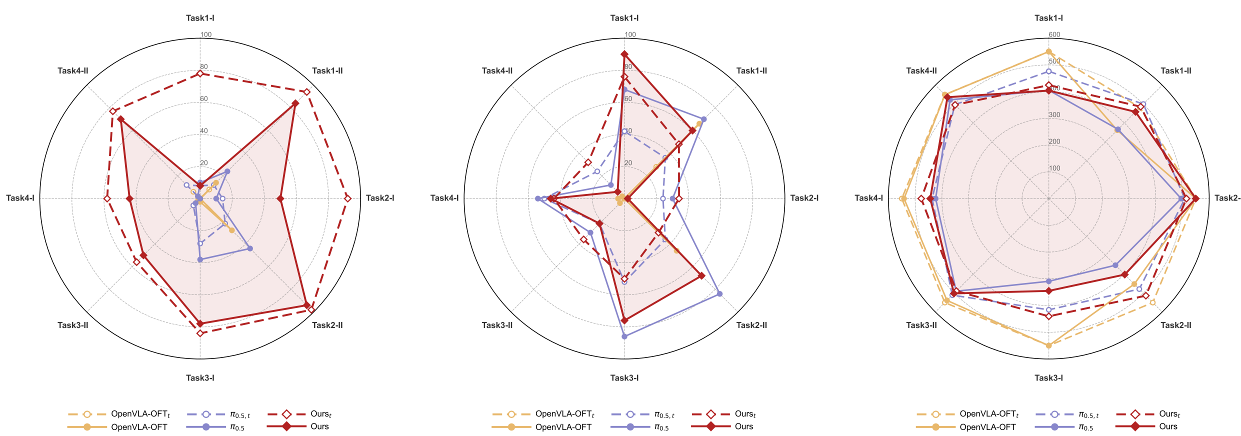
(a) Collision Avoidance Rate (↑) (b) Task Success Rate (↑) (c) Execution Time Steps (↓)

Figure S5: Performance on SafeLIBERO-Object suite. Comparison of CAR, TSR, and ETS.

Table S4: **Quantitative results on SafeLIBERO-Long suite.** Comparisons under both translational-only and full action space settings. **Bold** indicates the best result (higher is better for CAR/TSR, lower is better for ETS).

Task Description	**Level**	**Translational Action Space Only**									**Full Action Space**								
		OpenVLA-OFT$_t$			$\pi_{0.5,t}$			**Ours$_t$**			OpenVLA-OFT			$\pi_{0.5}$			**Ours**		
		C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓	C↑	T↑	E↓
Put both the alphabet soup and the cream cheese box in the basket	I	0.0	0.0	550.0	8.0	42.0	476.1	**78.0**	**76.0**	**424.4**	0.0	0.0	550.0	**10.0**	68.0	405.3	8.0	**90.0**	**404.0**
	II	8.0	28.0	**481.6**	12.0	36.0	500.0	**94.0**	**48.0**	486.0	14.0	66.0	**361.9**	24.0	**70.0**	366.3	**84.0**	60.0	458.8
Put both the alphabet soup and the tomato sauce in the basket	I	0.0	0.0	550.0	14.0	24.0	**508.9**	**92.0**	**34.0**	514.9	0.0	0.0	550.0	10.0	**30.0**	**495.2**	**50.0**	2.0	549.7
	II	22.0	0.0	550.0	22.0	**36.0**	**478.6**	**98.0**	30.0	513.5	28.0	46.0	452.1	44.0	**84.0**	**352.5**	**94.0**	68.0	401.8
Put the white mug on the left plate and put the yellow and white mug on the right plate	I	2.0	0.0	550.0	28.0	**52.0**	**416.0**	**84.0**	50.0	440.3	2.0	0.0	550.0	38.0	**86.0**	**309.3**	**78.0**	76.0	345.2
	II	6.0	0.0	550.0	6.0	22.0	510.7	**56.0**	**36.0**	**489.1**	2.0	4.0	538.7	4.0	**30.0**	**489.9**	**50.0**	22.0	501.7
Put the white mug on the plate and put the chocolate pudding to the right of the plate	I	0.0	0.0	550.0	0.0	**50.0**	**436.6**	**58.0**	44.0	477.1	0.0	4.0	541.5	0.0	**54.0**	**423.2**	**44.0**	46.0	443.5
	II	6.0	0.0	550.0	12.0	24.0	496.9	**77.0**	**32.0**	**495.7**	2.0	2.0	549.5	2.0	**12.0**	**522.4**	**70.0**	6.0	536.3
Average	-	5.5	3.5	541.5	12.8	35.8	**478.0**	**79.6**	**43.8**	480.1	6.0	15.3	511.7	16.5	**54.3**	**420.5**	**59.8**	46.3	455.1

C: CAR (Collision Avoidance Rate); **T**: TSR (Task Success Rate); **E**: ETS (Execution Time Steps).



(a) Collision Avoidance Rate (↑) (b) Task Success Rate (↑) (c) Execution Time Steps (↓)

Figure S6: Performance on SafeLIBERO-Long suite. Comparison of CAR, TSR, and ETS.